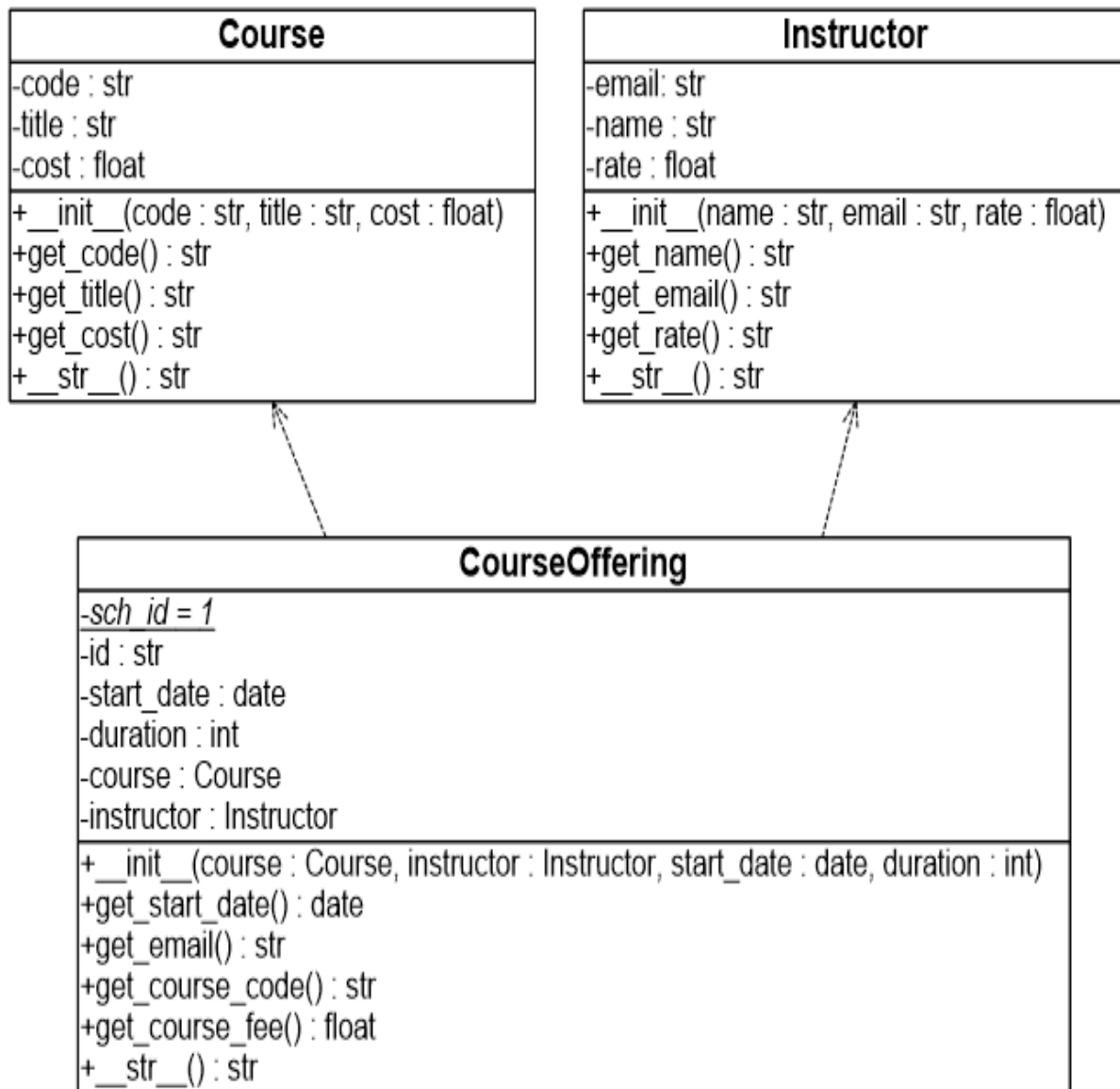


Question 1

Implement the following classes:



Course

code	Unique course code.
title	Course title.
cost	Cost to develop the course.
__str__	Include all attributes.
__eq__	Optional. Compare 2 objects based on course code.

Instructor

Email	Unique email.
Name	Instructor name.
Rate	Instructor rate per day.
__str__	Include all attributes.
__eq__	Optional. Compare 2 objects based on email.

CourseOffering

sch_id	Class variable, running number to be used as part of the identifier.
id	Unique identifier, combination of course code + sch_id E.g. Course code is 'IT101' sch_id is 1 Combination is IT101_1 Course code is 'BU121' sch_id is 2 (incremented) Combination is BU121_2
start_date	Instructor rate per day.
duration	Course duration, number of days.
Course	The course (an object) to be offered.
instructor	The instructor (an object) teaching the course.
get_course_fee	Course cost + instructor rate * duration
__str__	Include all attributes.
__eq__	Optional. Compare 2 objects based on email.

Implement the following functions:

`get_courses(filename : str) : list`

Read course data from a file, create Course objects, save into a list (or dictionary) and return the list (or dictionary). Do not include duplicate objects.

`get_instructors(filename : str) : list`

Read instructor data from a file, create Instructor objects, save into a list (or dictionary) and return the list (or dictionary). Do not include duplicate objects.

`get_course_offering(filename : str) : list`

Read course offering data from a file, create CourseOffering objects, save into a list (or dictionary) and return the list (or dictionary). Do not include duplicate objects.

main

- Create objects of the various classes, and test the methods.
- Call the `get_xxx` functions to create and test the objects.

Sample data

Course

```
IT101,Database,1200
IT102,Programming,1200
IT201,Web Application,1500
IT301,Big Data,1500
IT302,Project,800
```

Instructor

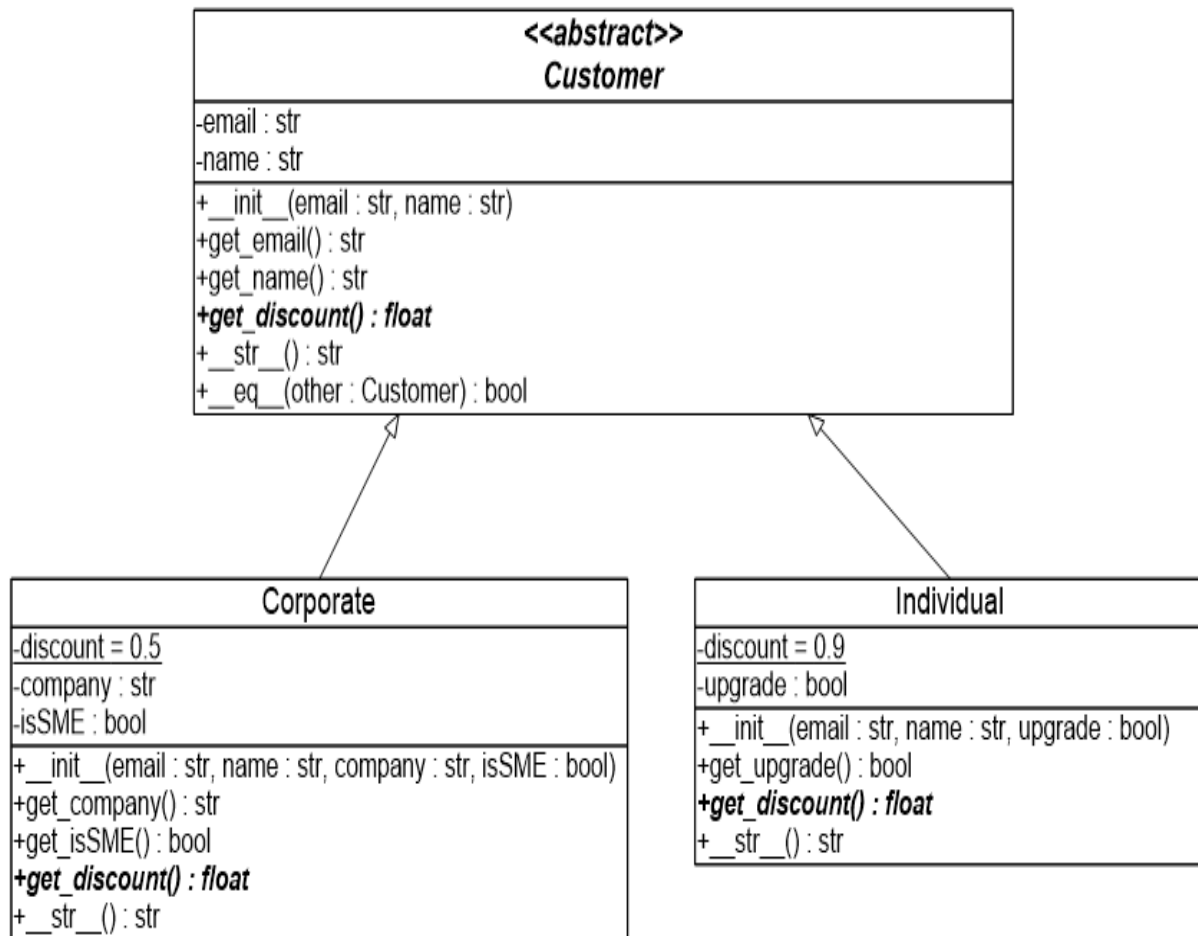
Wong TT,wongtt@mesg.com,300
Xing HM,xinghm@mesg.com,350
Low YB,lowyb@mesg.com,300

CourseOffering

IT101,wongtt@mesg.com,2026-01-10,2
IT102,xinghm@mesg.com,2026-01-12,1
IT201,lowyb@mesg.com,2026-02-20,2
IT301,lowyb@mesg.com,2026-01-25,2

Question 2

Implement the following classes:



Customer

email	Unique email.
name	Customer name.
__str__	Include all attributes.
__eq__	Optional. Compare 2 objects based on email.

Corporate

-discount	Class variable, discount for all corporate customers.
-company	Company name
-isSME	True if the company is a SME, False otherwise.
name	Customer name.
get_discount	50% discount as defined in the class variable. SME gets additional 10% (i.e. 60%) discounted.
__str__	Include all attributes (including attributes in base class)

Individual

-discount	Class variable, discount for customers receiving upgrade subsidy.
-upgrade	True if customer receiving upgrade subsidy.
get_discount	90% discount as defined in the class variable for customers receiving upgrade subsidy. 0% otherwise.
__str__	Include all attributes (including attributes in base class)

Implement the following functions:

get_customers(filename : str) : list

Read customer data from a file, create either Corporate or Individual objects, save into a list (or dictionary) and return the list (or dictionary). Do not include duplicate objects.

main

- Create objects of the various classes, and test the methods.
- Call the get_customers to create and test the objects.

Customer sample data

```
amy@mail.com,Amy,Y,
benison@mail.com,Benson,N
coby@mail.com,Coby,Power Pte Ltd,Y
don@mail.com,Don,First Bank,N
elton@mail.com,Elton,JCS,Y
frank@mail.com,Frank,Y
gary@mail.com,Gary,N
```

Q3.

Implement the following class. You may include additional attributes and/or methods.

- *add_xxx* methods return True if the operation is successful.
- *add_xxx* methods must not add duplicate object.
- write a load method (optional) to load data from the sample files (question 1 and 2).

TrainingCentre
-name : str -courses : dict -instructors : dict -o_courses : dict -customersr : dict
+__init__(name : Course) +add_course(code : str, title : str, cost : float) : bool +add_instructor(name : str, email : str, rate : float) : bool +add_course_offering(course : Course, instructor : Instruction, start_date : date, duration : int) : bool +add_customer(email : str, name : str, company : str, isSME : bool, upgrade : bool) : bool +__str__() : str